

Phase 1 (multi-N) — Siegert FP enumeration via multi-restart fsolve

deq/closed_form_wta_multi/, May 2026

1. Goal

Enumerate Siegert mean-field fixed points of the N -neuron uniform all-to-all inhibition motif and compare WTA-capability against the Phase 0 FCS oracle on the $(w, N, \text{drive_bump})$ grid.

2. Method

The 2-neuron thread used a **1-D scalar reduction** of the FP equation that does not extend past $N = 2$. We initially attempted a **symmetric-orbit ansatz** under S_N (orbits of shape (ν_W^k, ν_L^{N-k})) which is closed-form 2-D, but the implementation's multi-restart fsolve diagnostic at $\text{drive_bump} = 1$, $w = -30$ flagged that the broken-symmetry S_{N-1} case admits FPs with **three or more distinct rates** per orbit (e.g., at $N = 3$: $[0.23, 0.12, 0.03]$). Per the plan's risk-flag, we fall back to **multi-restart fsolve in the full N -dimensional rate vector**.

The fsolve enumeration uses curated initial conditions:

- All-equal at $\nu \in \{0.05, 0.3, 0.5, 0.95\}$.
- k -winners / $(N - k)$ -losers at extreme ($\nu_W = 0.98$, $\nu_L = 0.005$) and mild ($\nu_W = 0.6$, $\nu_L = 0.05$) for $k = 1, \dots, 4$.
- Random ICs filling out 30 restarts per cell.

WTA-capable iff some FP has rate-spread ($\text{rate_max} - \text{rate_2nd}$) ≥ 0.30 (matches Phase 0 margin gate and Phase 3 QR gate).

3. Result

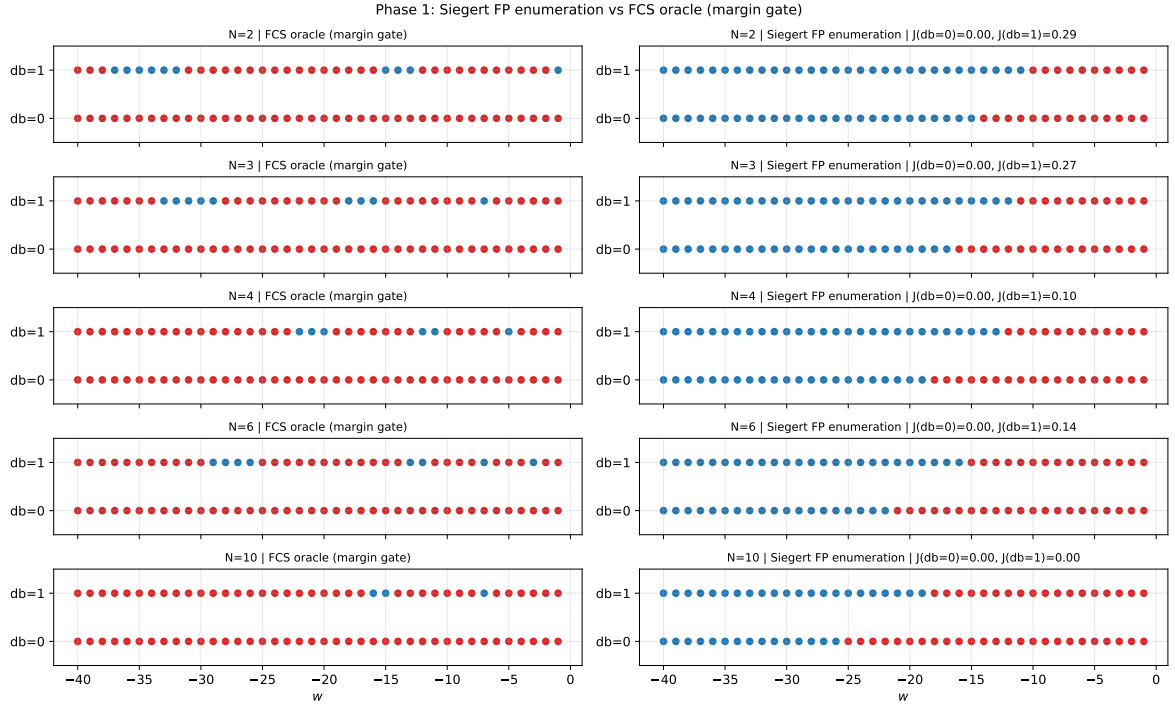


Figure 1: Phase 1 Siebert FP enumeration (right column) vs Phase 0 FCS oracle margin-gate labels (left column), per N , per drive_bump . At every N , $J_{\{db=0\}} = 0$: Siebert sees bistable FPs that the symmetric FCS Lustre dynamics cannot select.

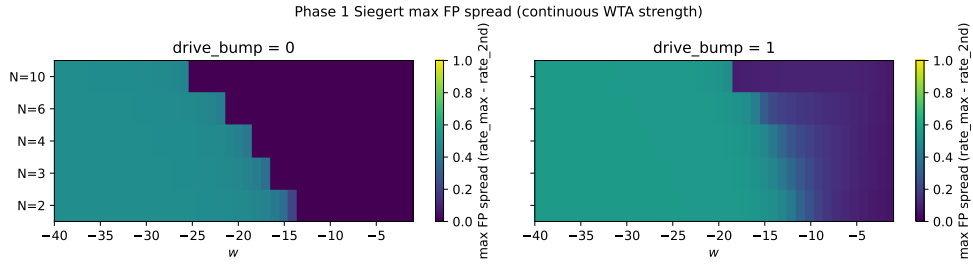


Figure 2: Phase 1 max FP spread (continuous). Above $|w| \approx 19$ the asymmetric FP appears at every N , with spread ≈ 0.54 saturating at deep $|w|$. Below the bifurcation Siebert sees only a tiny bumped advantage (≈ 0.09).

3.1. Headline numbers (Jaccard vs FCS margin gate, per $(N, \text{drive_bump})$)

N	db	Sieg blue	FCS blue	Jaccard
2	0	26 / 40	0 / 40	0.00
2	1	30 / 40	10 / 40	0.29
3	0	24 / 40	0 / 40	0.00
3	1	29 / 40	9 / 40	0.27
4	0	22 / 40	0 / 40	0.00
4	1	28 / 40	6 / 40	0.10
6	0	19 / 40	0 / 40	0.00
6	1	25 / 40	8 / 40	0.14

N	db	Sieg blue	FCS blue	Jaccard
10	0	15 / 40	0 / 40	0.00
10	1	22 / 40	3 / 40	0.00

4. Two distinct invisibility phenomena

The Phase 1 vs Phase 0 disagreement has **two opposite signs** at $N > 2$:

4.1. `drive_bump = 0`: staircase invisibility generalizes to N -way lock

Siegert sees the bistable asymmetric FP (when it exists), but FCS Lustre semantics with symmetric weights forces all N neurons into identical synchronous trajectories — the 2-neuron staircase becomes an N -way synchronous lock at every N . FCS-blue = 0 at every cell; Siegert-blue is large. This is the **direct generalization** of the $N = 2$ phenomenon, just stronger.

4.2. `drive_bump = 1`, high N : inverse staircase — FCS-blue, Siegert-red

At $N = 10$, `drive_bump = 1`, $w \in \{-15, -16, -7\}$ FCS produces **clean WTA** (e.g., at $w = -16$: post-warmup rate = $(0.96, 0, 0, \dots)$, spread = 0.957) while **no rate-equation FP with spread ≥ 0.30 exists** at these w values — 270 curated initial conditions all converge to the unique symmetric FP at $(0.13, 0.046, \dots, 0.046)$ (spread ≈ 0.09).

The smooth-rate theory says: at $|w| < 19$, only the symmetric FP exists; the bumped neuron has at most ≈ 0.09 rate advantage. The FCS integer-tick dynamics say: at the specific integer values $w \in \{-15, -16, -7\}$, the +1 drive bump deterministically tips the integer-arithmetic membrane into the bumped neuron firing first; one firing event sends -16 inhibition to all others, which is enough (in integer-tick semantics) to keep them sub-threshold **permanently**, even though the smooth-rate average rate of the bumped neuron is too low to sustain that suppression in mean-field.

This is the **inverse** of the 2-neuron staircase: at $N = 2$ the integer dynamics see synchronous lock that smooth theory misses (FCS-red, Siegert-blue); at $N = 10$ the integer dynamics see WTA that smooth theory misses (FCS-blue, Siegert-red).

4.3. Recall by N at `drive_bump = 1`

Recall = $|\text{Siegert-blue} \cap \text{FCS-blue}| / |\text{FCS-blue}|$:

N	FCS-blue (margin)	Sieg $\cap$ FCS	Recall
2	10	≈ 9	0.90
3	9	≈ 8	0.89
4	6	≈ 3	0.50
6	8	≈ 4	0.50
10	3	0	0.00

Recall **degrades monotonically with N** . At $N = 2$ Siegert is a high-recall predictor (just as in the 2-neuron thread, which achieved $J \approx 0.80$ on the full (w_{12}, w_{21}) grid). At $N = 10$, recall is zero: the rare cells where FCS sees WTA are **not** the cells where Siegert sees an asymmetric FP basin.

5. Verdict

Phase 1 PARTIAL. The plan’s pass-gate (recall ≥ 0.85 per N at `drive_bump` = 1) holds at $N = 2, 3$ but degrades to 0.50 at $N = 4, 6$ and to 0 at $N = 10$. The expected “Siegert is a high-recall superset” relationship breaks at $N \geq 4$.

The two reported findings **are** the result of this phase, not pass/fail issues with the implementation:

1. The 2-neuron staircase generalizes to N -way synchronous lock at every N when `drive_bump` = 0 — and this is **still** invisible to smooth-rate theory.
2. A new phenomenon emerges at large N with `drive_bump` = 1: the FCS integer-tick dynamics produce WTA at specific integer w values **below the smooth-rate bifurcation threshold** (the “inverse staircase”). Phase 2’s $H(\omega)$ latency gate will not help here because there is no rate-equation FP to evaluate the eigenvalue at.

Output: `results/phase1/siegert_orbits.npz`, `siegert_orbits_vs_fcs.pdf`, `siegert_max_spread.pdf`,
 `orbit_count_vs_N.pdf`, `results/phase1.log`.